



SIMATS ENGINEERING
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Automata-Based Intelligent ATS Resume Analysis and Optimization System

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ABSTRACT

- ATS automatically filters resumes before recruiters review them.
- Many qualified candidates are rejected due to formatting and structural issues.
- This system analyzes resumes using Theory of Computation concepts.
- It validates resume content, suggests improvements, and generates ATS compatibility scores.
- Provides job-role-based recommendations to improve employability.

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PROBLEM STATEMENT

- Many companies use Applicant Tracking Systems (ATS) to automatically screen resumes before they reach recruiters.
- Qualified candidates are often rejected due to poor formatting, missing keywords, invalid patterns, and improper resume structure.
- Existing resume analyzers provide only basic scores with limited explanations and improvement suggestions.
- There is a need for an intelligent system that validates, optimizes, and evaluates resumes using Theory of Computation concepts.

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OBJECTIVES

- To extract and analyse resume content using Lexical Analysis.
- To validate resume information using Regular Expressions and Deterministic Finite Automata (DFA).
- To optimize resume structure and recommend improvements using Context-Free Grammar (CFG).
- To generate ATS compatibility and job matching scores for selected job roles.

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Automata-Based ATS Resume Analysis and Optimization System



MODULE 1: TECHNICAL DETAILS

- Accepts resumes in PDF and DOCX formats. Extracts raw text using document parsing libraries.
- Performs lexical analysis by tokenizing the text into meaningful units.
- Identifies resume sections such as Name, Education, Skills, Projects, Experience, Certifications, and Contact Details.
- Converts unstructured resume content into a structured JSON/database format.
- Passes the parsed data to the Validation Engine.

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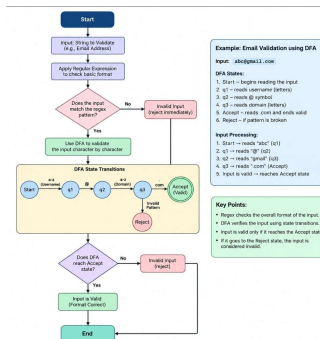
MODULE 2: RESUME VALIDATION ENGINE

Purpose
Validate extracted information.
Detect formatting errors.
Check mandatory fields.
Validation
Email
Phone
LinkedIn
GitHub
Dates
Missing sections
TOC Concepts
Regular Expressions
Deterministic Finite Automata (DFA)



MODULE 2: TECHNICAL DETAILS

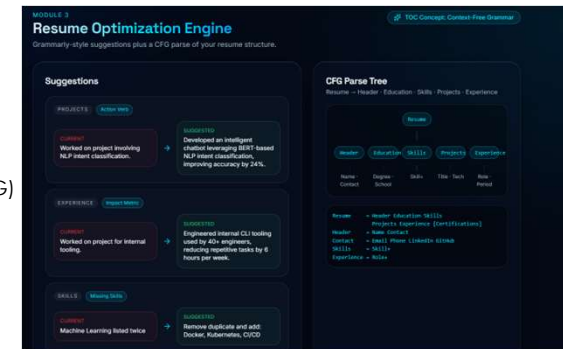
- Receives structured data from the Resume Parsing Engine.
- Uses Regular Expressions (Regex) to validate email addresses, phone numbers, LinkedIn URLs, GitHub links, and date formats.
- Applies Deterministic Finite Automata (DFA) to verify input patterns through state transitions.
- Detects missing mandatory sections, duplicate entries, and formatting inconsistencies.
- Generates a comprehensive validation report highlighting errors and warnings.
- Sends validated data to the Optimization Engine.



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MODULE 3: RESUME OPTIMIZATION ENGINE

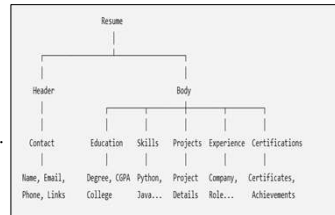
Purpose
Suggest corrections.
Improve ATS compatibility.
Recommend missing skills.
Improve formatting.
TOC Concepts
Context-Free Grammar (CFG)
Parse Tree



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MODULE 3: TECHNICAL DETAILS

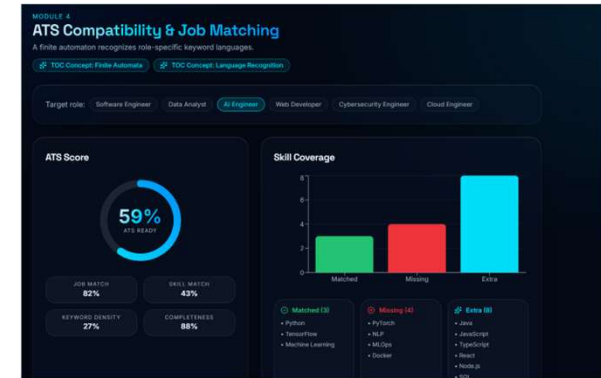
- Analyzes the logical structure of the resume using Context-Free Grammar (CFG).
- Verifies the correct sequence of sections such as Header, Education, Skills, Projects, and Experience.
- Suggests ATS-friendly formatting improvements and section reordering. Recommends stronger action verbs, relevant keywords, and missing technical skills.
- Generates optimized resume recommendations to improve readability and ATS compatibility.
- Transfers the optimized resume to the ATS Scoring Engine.



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MODULE 4: ATS COMPATIBILITY & JOB MATCHING

Purpose
Compare resume with selected job role.
Generate ATS Score.
Calculate Job Match Score.
TOC Concepts
Finite Automata
Language Recognition
Regular Expressions



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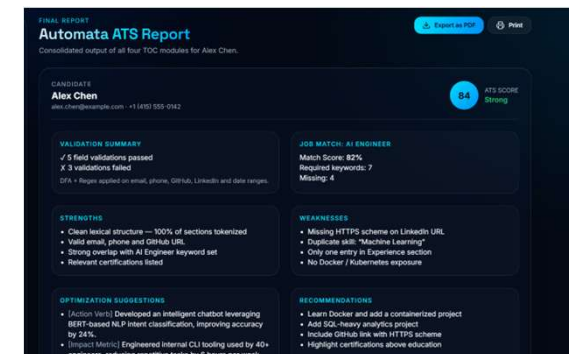
MODULE 4: TECHNICAL DETAILS

- Compares the optimized resume with the requirements of a selected job role.
- Uses Finite Automata and Language Recognition principles to identify matching and missing skills.
- Calculates ATS Compatibility Score based on skill match, resume completeness, validation status, and keyword relevance.
- Generates Job Match Score and identifies missing competencies.
- Recommends certifications, technical skills, and improvements to increase employability.
- Produces a downloadable ATS analysis report with scores and recommendations.

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TESTING & EVALUATION

Test Cases
Different resume formats.
Missing sections.
Invalid emails.
Invalid phone numbers.
ATS score comparison.
Metrics
Accuracy
Validation Success Rate
Processing Time
ATS Compatibility Score



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SDG CONTRIBUTION



SDG 4:

Quality Education
Helps students prepare professional resumes.

SDG 8:

Decent Work and Economic Growth
Improves employability and job readiness.



CONCLUSION & FUTURE WORK



Conclusion:

Developed an intelligent ATS resume analyzer.
Applied TOC concepts for validation and optimization.
Improved resume quality and ATS compatibility.

Future Work:

AI-based interview preparation.
Multi-language resume support.
Real-time recruiter feedback.
Integration with job portals.

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*Thank
you!*